

```

    bound1 = tree[0][1]
    bound2 = tree[0][2]
    bound3 = tree[1][2]
    if observation.loc[variable_tocheck] < bound2:
        tree = tree[0][3]
    else:
        tree = tree[1][3]
    if isinstance(tree, float):
        keepgoing = False
        prediction = tree
    return(prediction)

```

Next, we can create a loop that goes through any portion of our dataset and gets any tree's happiness prediction for that portion. In this case, let's try a tree with a maximum depth of four:

```

predictions=[]
outcome_variable = 'happy'
maxdepth = 4
thetree = getsplit(0,ess,variables,outcome_variable)
for k in range(0,30):
    observation = ess.loc[k,:]
    predictions.append(get_prediction(observation,thetree))

print(predictions)

```

This code just repeatedly calls the `get_prediction()` function and appends the result to our predictions list. In this case, we made predictions only for the first 30 observations.

Finally, we can check how these predictions compare to the actual happiness ratings, to see what our total error rate is. Here, we'll make predictions for our entire dataset, and calculate the absolute differences between our predictions and the recorded happiness values:

```

predictions = []

for k in range(0,len(ess.index)):
    observation = ess.loc[k,:]
    predictions.append(get_prediction(observation,thetree))

ess.loc[:,'predicted'] = predictions

```